



Estd:1946

THE NATIONAL INSTITUTE OF ENGINEERING

(An Autonomous Institution under VTU, Belagavi)
Koorgalli Village, Hootagalli Industrial Area, Mysuru -570018,
Department of Computer Science and Engineering

Course Plan for the Term ODD 2026-27

Course Instructor: Dr. PADMINI.M.S
Course Title: Data Structures and Applications
L:T:P:TW/SL - 42:0:0:48

Term: 10/08/2026 to 28/11/2026
Course Code: 1BCS304
Semester: 3 CSE B

LESSON PLAN

Session No.	Planned Date	Content	Actual Date/s	Mode of Delivery (Classroom/Online)	Link for online content	Link for Course Content	Comments , if any by CI/HoD
1.	11.08.26	Instructor information, Course Information, Relevance of Course, Course Resources, Evaluation pattern, and OBE Awareness Introduction to Data Structures		PPT	https://www.coursera.org/learn/data-structures/supplement/G4YP1/welcome	Moodle Link: https://moodlegurukul.nie.ac.in/course/view.php?id=7532#section-1	
2.	12.08.26	Classification of data structures, Operations on data structures, Abstract data type		PPT + Chalk & Talk	https://www.geeksforgeeks.org/dsa/what-is-data-structure-types-classifications-and-applications/		
3.	14.08.26	Stacks: Introduction, Array Representation of Stacks,		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/UdKzQ/stacks https://www.geeksforgeeks.org/dsa/stack-data-structure/		
4.	14.08.26	Linked Representation of Stacks		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/UdKzQ/stacks https://www.geeksforgeeks.org/dsa/implement-a-stack-using-singly-linked-list/		

5.	18.08.26	Operations on a Stack - (push, pop, peek, display)		Demonstration + Chalk & Talk	https://www.coursera.org/learn/data-structures/lecture/UdKzQ/stacks https://www.geeksforgeeks.org/dsa/implement-a-stack-using-singly-linked-list/		
6.	19.08.26	Applications of Stacks- Conversion of an infix expression into a postfix expression		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/applications-advantages-and-disadvantages-of-stack/		
7.	21.08.26	Evaluation of a postfix expression		Demonstration	https://www.geeksforgeeks.org/dsa/evaluation-of-postfix-expression/		
8.	21.08.26	Recursion- Factorial of N		PPT + Chalk & Talk	https://www.geeksforgeeks.org/dsa/program-for-factorial-of-a-number/		
9.	25.08.26	Tower of Hanoi		Demonstration	https://www.geeksforgeeks.org/dsa/c-program-for-tower-of-hanoi/		
10.	28.08.26	Queues: Introduction		PPT	https://www.coursera.org/learn/data-structures/lecture/EShpq/queues	Moodle Link: https://moodlegurukul.nie.ac.in/course/view.php?id=7532#section-2	
11.	28.08.26	Array Representation of Queues		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/EShpq/queues		
12.	8.09.26	Operations on Queues		Demonstration + Chalk & Talk	https://www.coursera.org/learn/data-structures/lecture/EShpq/queues		
13.	9.09.26	Linked Representation of Queues		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/EShpq/queues		
14.	11.09.26	Types Of Queues: Circular Queue		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/different-types-of-queues-and-its-applications/		
15.	11.09.26	Deque		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/different-types-of-queues-and-its-applications/		
16.	15.09.26	Priority Queues		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/different-types-of-queues-and-its-applications/		
17.	16.09.26	Applications of Queues		Flipped Classroom + Discussion	https://www.geeksforgeeks.org/dsa/different-types-of-queues-and-its-applications/		
18.	18.09.26	Linked Lists – Introduction		PPT	https://www.coursera.org/learn/data-structures/lecture/kHhgK/singly-linked-lists		

19.	18.09.26	Singly linked lists -Traversing, Searching, Insertion, Deletion, Display		Demonstration + Chalk & Talk	https://www.coursera.org/learn/data-structures/lecture/kHhgK/singly-linked-lists	Moodle Link: https://moodlegurukul.nie.ac.in/course/view.php?id=7532#section-3	
20.	22.09.26	Circular linked lists - Inserting a Node in a Circular Linked List		Demonstration	https://www.coursera.org/learn/data-structures/lecture/jpGKD/doubly-linked-lists		
21.	29.09.26	Circular linked lists - Deleting a Node in a Circular Linked List		Demonstration	https://www.coursera.org/learn/data-structures/lecture/jpGKD/doubly-linked-lists		
22.	30.09.26	Doubly linked lists - Traversing, Searching		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/jpGKD/doubly-linked-lists		
23.	6.10.26	Doubly linked lists - Insertion, Deletion, Display.		Demonstration	https://www.coursera.org/learn/data-structures/lecture/jpGKD/doubly-linked-lists		
24.	7.10.26	Circular doubly linked lists - Insert Begin, Insert End		Demonstration	https://www.geeksforgeeks.org/dsa/circular-linked-list/		
25.	9.10.26	Delete Front, Delete End, Display		Demonstration	https://www.geeksforgeeks.org/dsa/circular-linked-list/		
26.	9.10.26	Applications of Linked Lists		Flipped Classroom + Discussion	https://www.geeksforgeeks.org/dsa/applications-of-linked-list-data-structure/	Moodle Link: https://moodlegurukul.nie.ac.in/course/view.php?id=7532#section-4	
27.	13.10.26	Trees: Introduction, Types of Trees		PPT	https://www.coursera.org/learn/data-structures/lecture/95qda/trees		
28.	14.10.26	Binary Tree Traversing: pre-order, in-order, post-order, level-order		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/fr51b/tree-traversal		
29.	16.10.26	Binary Search Trees - Operations on Binary Search Trees		Demonstration	https://www.coursera.org/learn/data-structures/lecture/95qda/trees https://www.geeksforgeeks.org/dsa/binary-tree-traversal/		
30.	16.10.26	Threaded Binary Trees: One-way Threading		PPT + Chalk & Talk	https://www.coursera.org/learn/data-structures/lecture/95qda/trees https://www.geeksforgeeks.org/dsa/threaded-binary-tree/		
31.	23.10.26	Balanced Trees: Splay Trees		PPT + Demonstration	https://www.coursera.org/learn/data-structures/lecture/SGm2x/splay-trees-introduction https://www.geeksforgeeks.org/dsa/introduction-to-splay-tree-data-structure/		

32.	27.10.26	Multiway Search trees – B trees		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/introduction-of-b-tree-2/		
33.	28.10.26	B+ trees and Trie		PPT + Demonstration	https://www.geeksforgeeks.org/dbms/introduction-of-b-tree/ https://www.geeksforgeeks.org/dsa/trie-insert-and-search/		
34.	30.10.26	Applications of trees		Flipped Classroom + Discussion	https://www.coursera.org/learn/data-structures/programming/RSvTH/programming-assignment-4-binary-search-trees https://www.geeksforgeeks.org/dsa/applications-of-tree-data-structure/		
35.	3.11.26	Sorting: Introduction to Sorting, Radix Sort		PPT + Demonstration	https://www.geeksforgeeks.org/dsa/radix-sort/	Moodle Link: https://moodlegurukul.nie.ac.in/course/view.php?id=7532#section-5	
36.	4.11.26	Tree Sort		Demonstration	https://www.geeksforgeeks.org/dsa/tree-sort/		
37.	6.11.26	Hashing: Introduction, Hash Tables		PPT	https://www.coursera.org/learn/data-structures/lecture/5e8QH/hash-tables		
38.	11.11.26	Different Hash Functions		PPT + Chalk & Talk	https://www.coursera.org/learn/data-structures/lecture/vOqnE/hash-functions		
39.	13.11.26	Collisions: Collision Resolution by Open Addressing		Demonstration	https://www.coursera.org/learn/data-structures/lecture/fYKm7/direct-addressing		
40.	20.11.26	Collision Resolution by Chaining		Demonstration	https://www.coursera.org/learn/data-structures/lecture/2yY2h/chaining		
41.	24.11.26	Pros and Cons of Hashing		PPT + Discussion	https://www.coursera.org/learn/data-structures/lecture/dWNAc/chaining-implementation-and-analysis		
42.	25.11.26	Applications of Hashing		Flipped Classroom + Demonstration	https://www.coursera.org/learn/data-structures/lecture/e4vuN/applications-of-hashing		

Textbooks:

1. Reema Thareja, Data Structures using C, 2nd Edition, 2018, Oxford University Press.
2. Aaron M Tenenbaum, Yedidyah Langsam and Moshe J Augenstein, Data Structures using C, 2014, low price edition, Pearson education.

Reference Books:

1. Richar F Gilberg and Behronz A Forouzan, Data Structures, A Pseudocode Approach with C”, 2nd Edition, 2012, Thomson.
2. Horowitz, Sahni, Anderson-Freed, Fundamentals of Data Structures in C, 2nd Edition, 2011, Universities Press.

Online Resources:

1. NPTEL: Data Structures and Algorithms Design - https://onlinecourses.nptel.ac.in/noc25_cs81/preview
2. Coursera: Data Structures - <https://www.coursera.org/programs/faculty-learning-program-iqr5x/learn/data-structures>

Evaluation Pattern

BT Level	Test 1 (25)	Test-2 (25)	Quiz (20)	Experiential Learning (30)	SEE (100)
L2 (Understand)	10	8	0	2	30
L3 (Apply)	10	10	10	8	40
L4 (Analyse)	5	7	10	20	30
Total	25	25	20	30	100

Evaluation Pattern for Experiential Learning (30 Marks)

Component	Evaluation Criteria	Marks
CodeChef/ Case Studies: 30 marks Apply Data Structures concepts to solve real-world problems through analysis, design, implementation, testing, and evaluation of software solutions.	Proposal Submission (Problem Definition + DS Selection)	5
	Design & Algorithm Development	5
	Coding & Implementation	10
	Testing, Analysis & Report	5
	Demonstration & Viva	5
Suggested Tools/ AI assisted Tools: Python Tutor, GitHub Co-pilot, Claude AI		

Signature of Course instructor

Signature of Course Coordinator/Faculty Mentor

Signature of HoD